

No Previews in Pyongyang

Reform discourse and policy change in North Korea, 1987–2020

Steven Denney · Leiden University

Peter Ward · Sejong Institute

ECNK Webinar

European Centre for North Korean Studies, University of Vienna

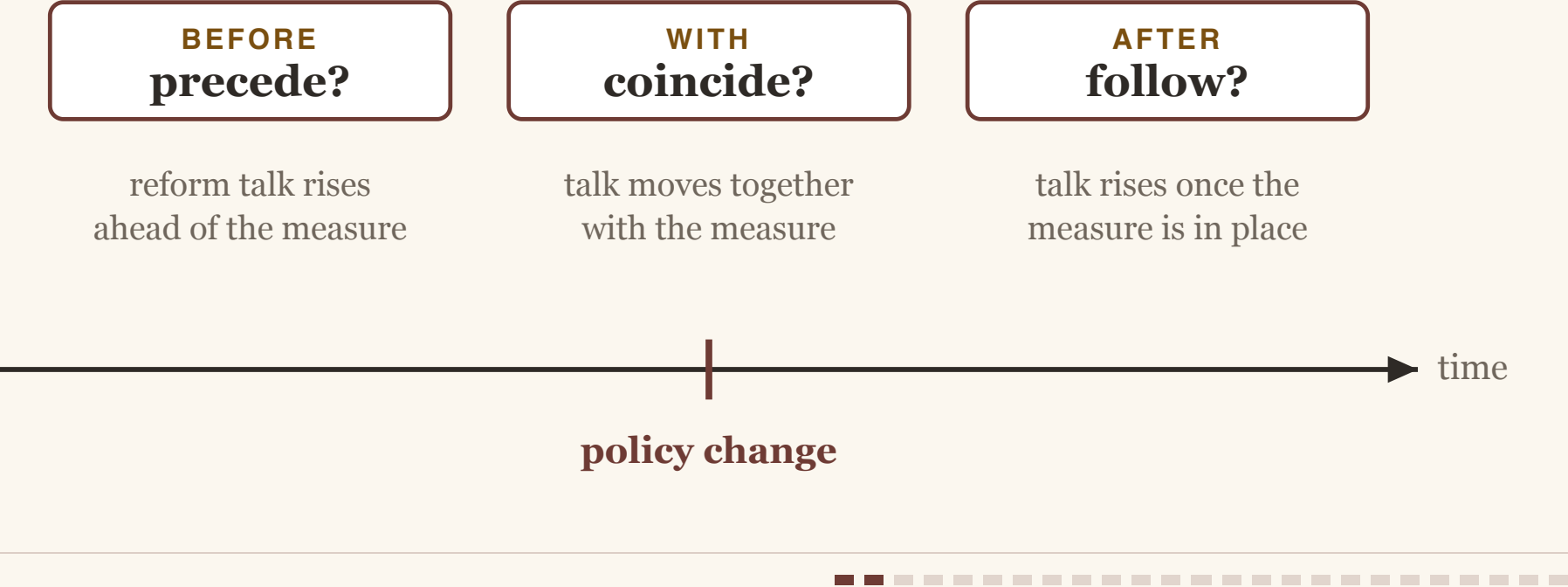
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THE QUESTION

In official North Korean sources, does reform talk precede, coincide with, or follow policy?

Everyone agrees the economy has been reformed repeatedly. What is disputed is how policy is made in Pyongyang, and what state propaganda does in that process.



Why it matters

Comparativists see two kinds of propaganda

Soft propaganda persuades and deceives. Hard propaganda intimidates and deters unrest. The policy function has been largely ignored.

e.g. Carter and Carter 2023 · Huang 2015

North Korea watchers read it for policy

Agitprop monopolizes the information environment, so analysts mine it for policy, elite dynamics, and the state of the country. Some see options being debated between party, cabinet, and military.

e.g. Carlin and Wit 2006 · McEachern 2009

The timing has never been tested

Prior quantitative work noted shifts in topics and tone, but read titles rather than whole articles, and never dated the response.

e.g. Choi and Lecy 2012 · Hastings and Lee 2023

What theory expects

EXISTING THEORY

Theory 1: Legitimation
Regime decides internally in a closed fashion and policy content is not provided, just praise for the new policy and slogans.
Carter and Carter 2023

Theory 2: Intimidation
Regime uses propaganda to project strength and scare would-be dissenters without actual policy content.
Huang 2015

Theory 3: Policy content
The text does carry real policy content.
Carlin and Wit 2006 · McEachern 2009

OUR EXPECTATIONS

Precede
The text airs reformist ideas before the decision.

Coincide
The text arrives with the decision, aligning cadres and readers.

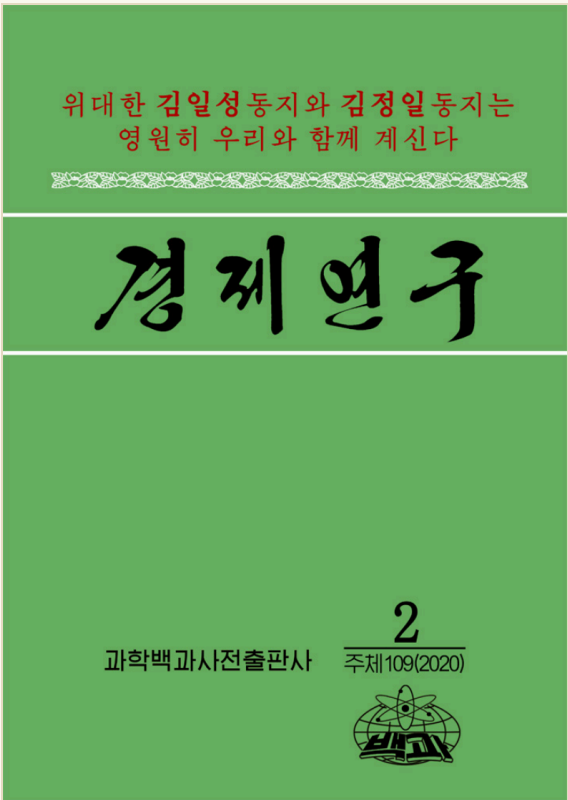
Follow
The text explains the decision after it is taken.



Sources



The journal



Cover of the 2020-2 issue.

Kyŏngje Yŏngu (경제연구)

The regime’s economics quarterly, written for planners, managers, officials and academics. Paeans to great leadership sit alongside technical discussion of production and futures contracts.

Yang 2009

Articles covering the period 1987 to 2020

3,036 articles, full text in DPRK orthography. Four issues a year, on a fixed schedule.

What is missing

Three quarterly issues (1995-4, 2019-4, and 2020-3) were unavailable, leaving 133 of 136 issues. In addition, 82 articles contain only a title and author byline, without any body text.



Data and methods



Building the measure

1

Prepare

The full text of all 3,036 articles, 1987 to 2020, in the North's own spelling and vocabulary.

2

Seed

Our own list first: 55 reform terms from close reading, plus a matching retrenchment list. Machines only extend it.

3

Discover

An LLM, EXAONE 3.5 on the Leiden HPC, labels every article without seeing our list. Its disagreements with the list show where missing vocabulary hides.

4

Embed

No LLM here: a small statistical model of the journal maps which words keep company with our seed terms. Look-alikes become candidates.

5

Validate

Three agents built on a frontier LLM, separate from EXAONE, judge each candidate against our reform definition, six mechanisms that loosen central control. Of 32 candidates, 3 are added to the lexicon and 11 to an extended list used for robustness.

6

Measure

An expert review fixes the lexicon at 40 terms. Distinct reform terms per article, minus retrenchment terms, averaged per quarter: the outcome variable.

The reform list

40 anchor terms, drawn from close reading. Two obvious examples from three of the six mechanisms.

Enterprise autonomy

경영권 (management rights)

계획권 (planning rights)

Price and marketization

수요와공급 (supply and demand)

농민시장 (farmers' market)

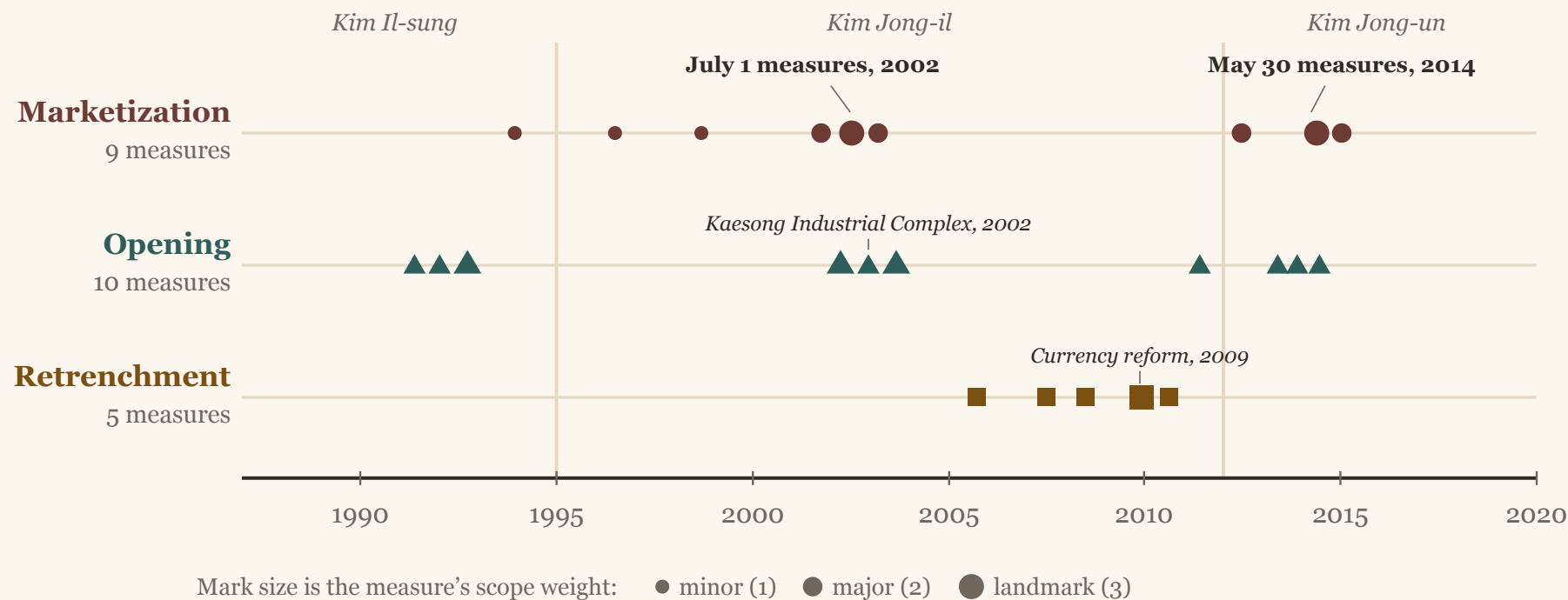
Incentives and distribution

번수입 (earned income)

물질적자극 (material incentives)



The policy timeline

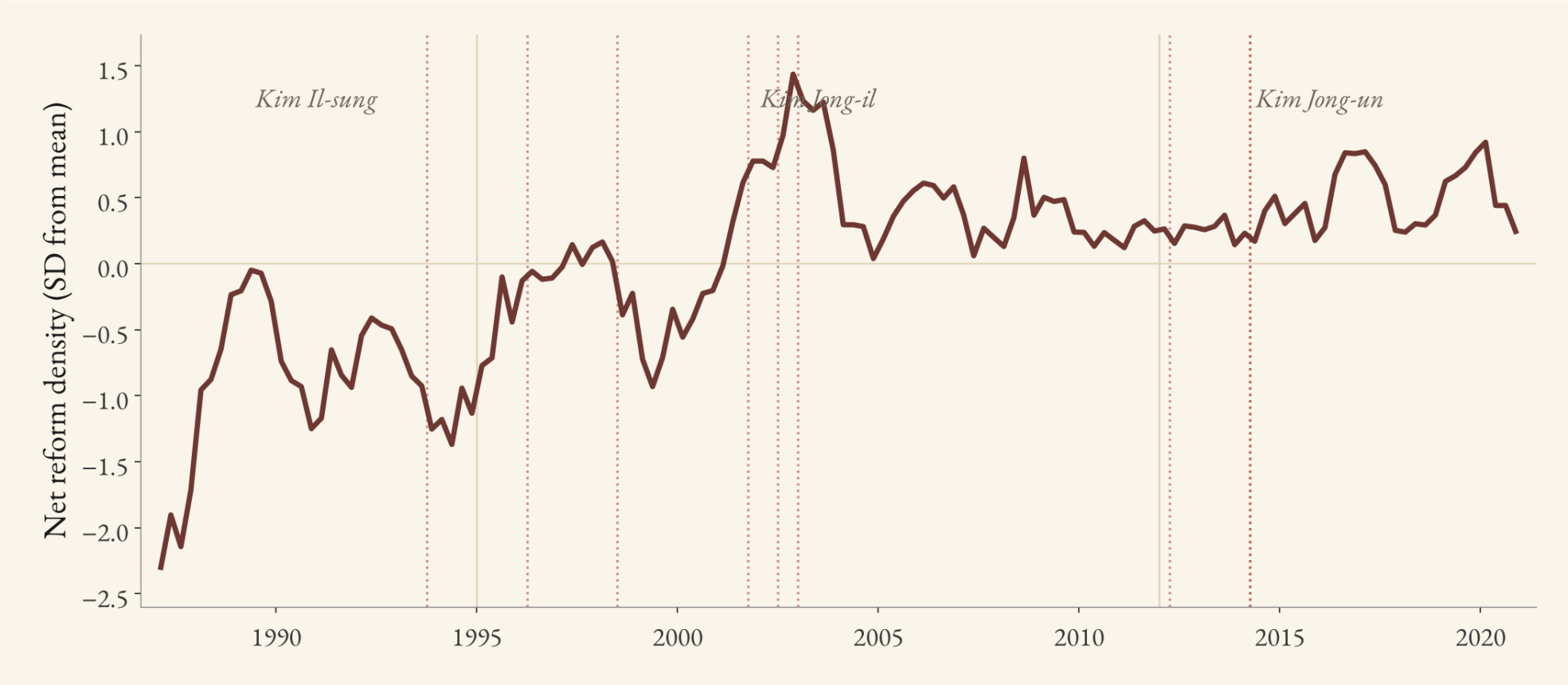


The policy events tested in the analysis. Larger marks indicate broader reforms.



The time series

Reform talk over time. The red lines mark the nine marketization measures.

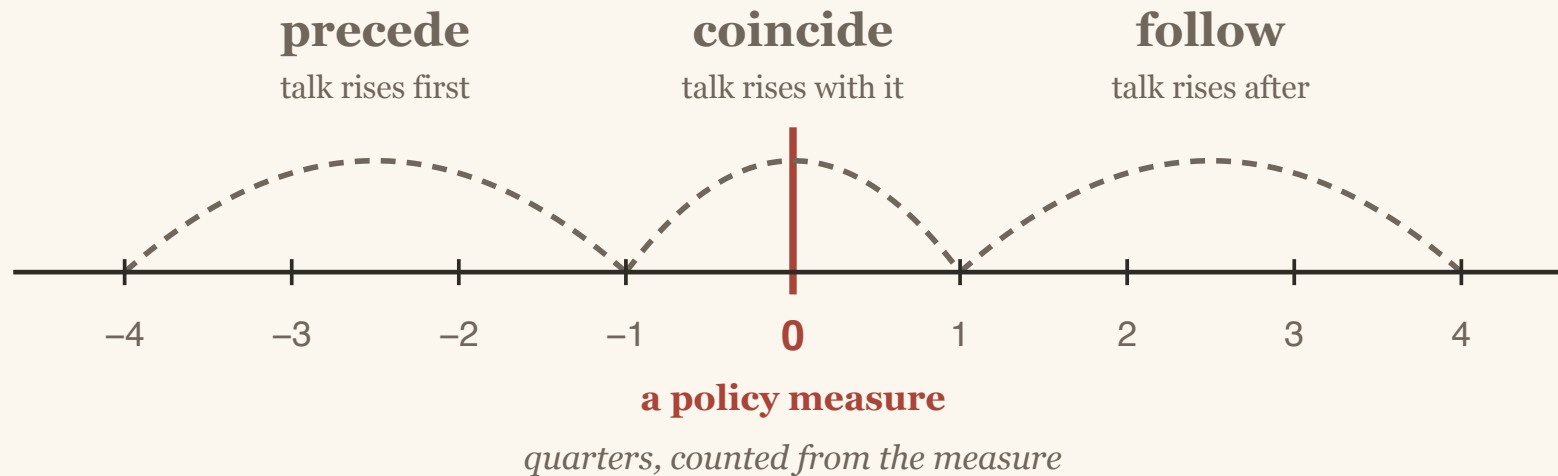


This is the outcome the estimator tries to explain.



The test

Three patterns are possible. Each one would look different on the calendar, so the data can tell them apart.



We estimate the response quarter by quarter to see whether reform talk comes before, with, or after policy.



The estimator

We estimate whether policy is followed by increases in reform talk, after accounting for persistence, era effects, and trends.

future reform talk (h quarters later) = $\alpha + \beta_h \cdot \text{policy}_t + \gamma \cdot \text{reform talk}_{t-1} + \text{era} + \text{trend} + \text{error}$

β_h is what we want to estimate
For each horizon h , β_h tells us how much reform talk increases after a policy measure, beyond what we would otherwise expect. We estimate one β for each horizon, $h = 0, \dots, 8$.

The gray terms explain the baseline
They capture how much reform talk we would expect even without a policy measure: persistence from last quarter, differences across eras, and long-run trends. β_h measures only what is left over.

The same test runs backward
Directions swapped: does reform talk rise *before* policy? A preview would show here.

Two choices were made before we ran anything
Bigger measures count for more, weighted 1 to 3 by scope. We look within the first year after each measure.



Findings



Discourse follows, never leads



+0.29 Three quarters after a measure, the journal runs about a third of a standard deviation above its usual level. In plain terms: in an issue of roughly forty articles, about **three or four of them pick up one more reform term** than they otherwise would. Small per article, visible per issue, and gone by the next quarter.

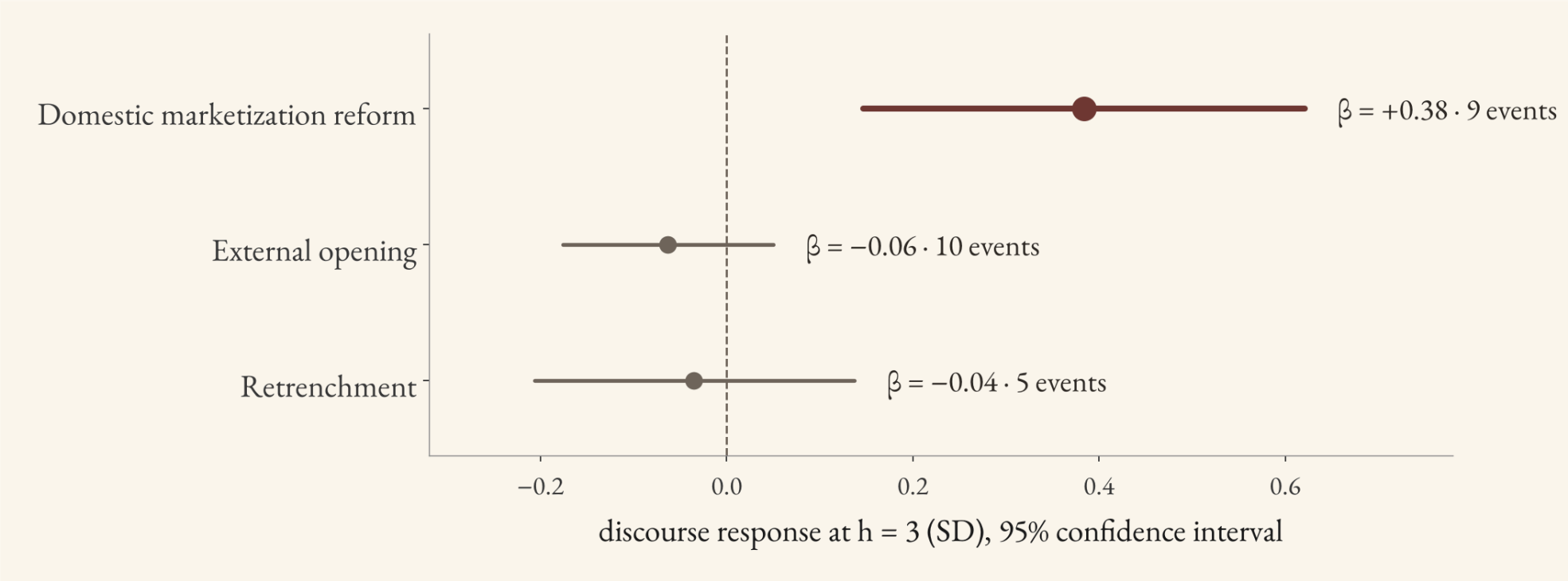
The solid line shows the estimated response after policy. The dashed line tests whether reform talk appears before policy. The smaller rise at h = 6 is the clustered 2001-03 measures seen twice, not a second response (appendix).



FINDINGS

Only marketization responds

Only domestic marketization is followed by more reform talk.



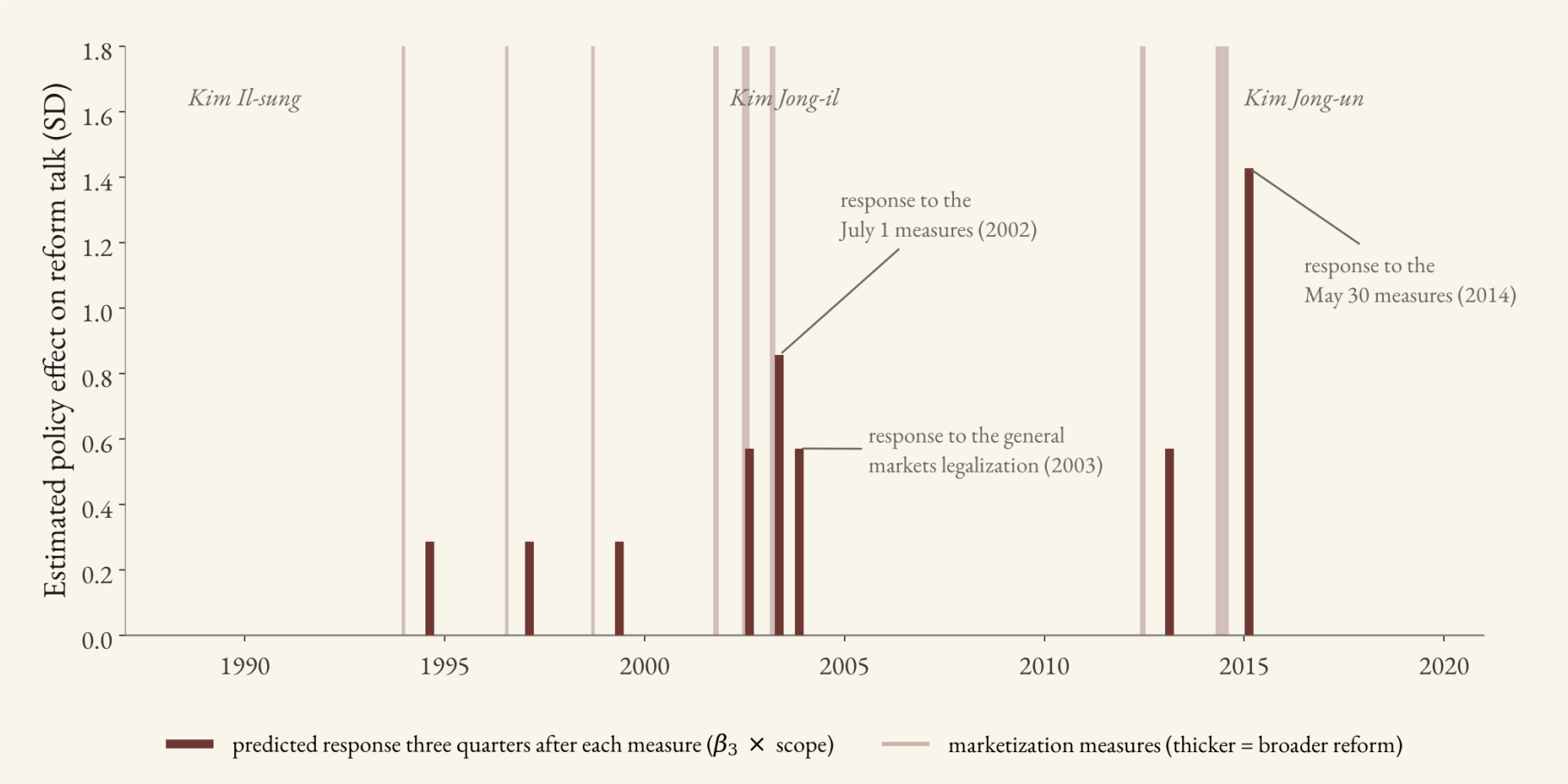
Only the marketization estimate is clearly above zero.



FINDINGS

Where the estimated effects appear

The predicted rise in reform talk, three quarters after each of the nine measures.



Predicted responses from the estimator, not observed reform talk.



FINDINGS

Why three quarters is plausible



The publication cycle naturally predicts a response about three quarters after policy.



Reading the text



Text as data

What are the post-policy articles actually doing?

Pool the post-policy articles

Every article published in the year after each of the nine measures, kept if it is actually about reform: at least two reform terms, more reform than retrenchment. 130 articles.

Count three functions, by machine

A script counts words that justify the policy, explain how it works, or urge action on it, compared against the journal's ordinary reform writing. Nothing is tallied by hand.

Read three cases closely

The biggest measures, at the centers of the two response clusters:

1. The July 1 measures, 2002
2. General markets legalized, 2003
3. The May 30 measures, 2014

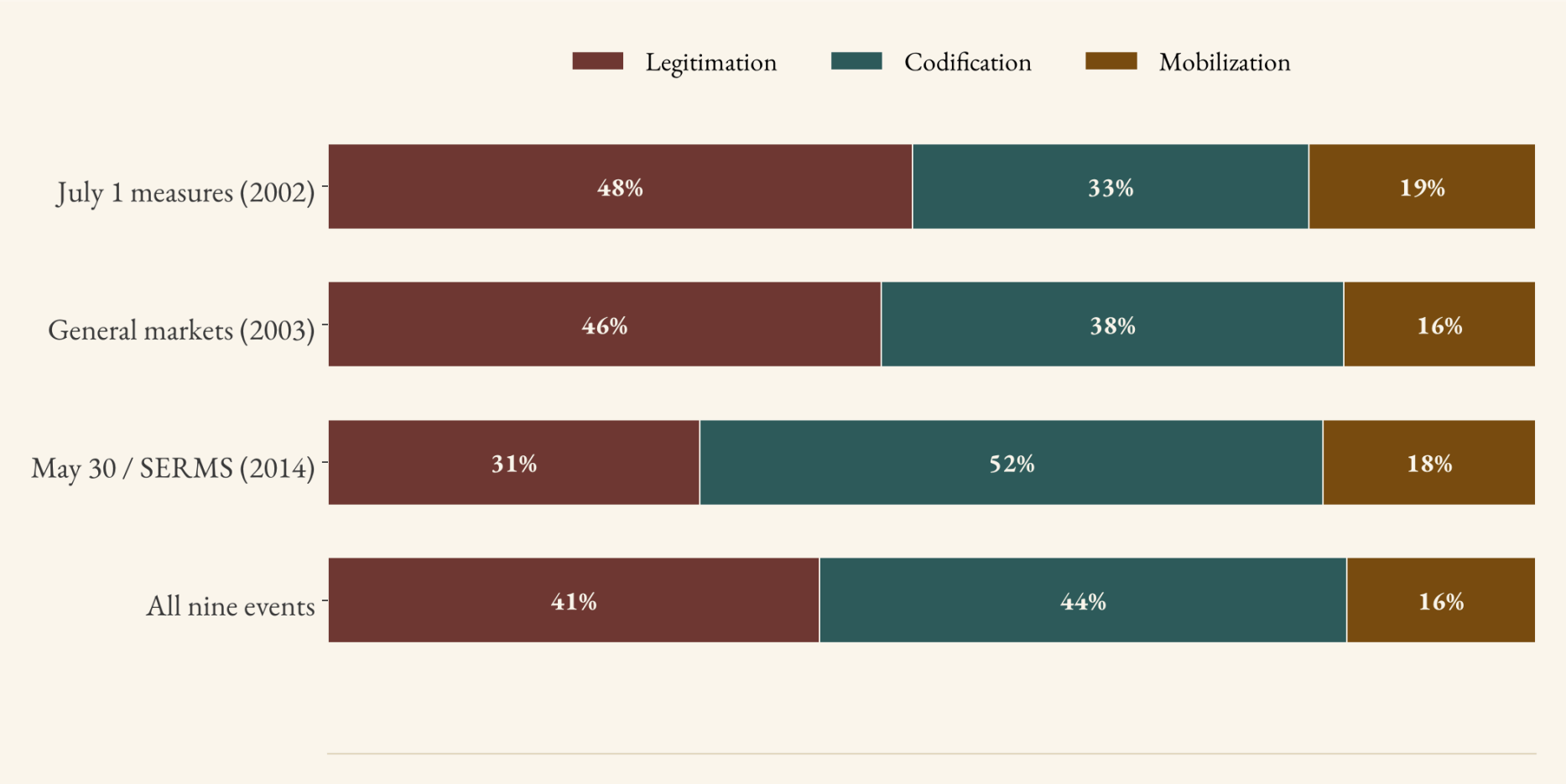
- This is our **initial, human read** of what the post-policy text says.
- **Next: a machine check.** An LLM classifies the domestic-marketization content against the same categories. Done transparently and fairly, it removes our bias as readers.



FINDINGS

Framing

Most post-policy reform talk explains or justifies policy rather than mobilizing readers.



In the journal’s own words

Examples from the post-policy response identified by the estimator, both from the year after the 2003 legalization of general markets.

JUSTIFYING IT · 경제연구 2003, NO. 4

“경애하는 장군님께서는 최근 번수입에 의한 경영방법을 받아들여 모든 독립채산제기업소들에서 경영손실을 없애고 실리를 보장할데 대한 문제 … 조치를 취해주시었다.”

The respected General has recently adopted the earned-income management method and taken measures to eliminate losses and guarantee practical benefit at every independent-accounting enterprise.

EXPLAINING HOW IT WORKS · 경제연구 2004, NO. 1

“현 시기 지방별 물자교류시장을 합리적으로 조직하고 그것을 잘 리용하는것은 지방공업공장들의 생산을 정상화하기 위한 중요한 요구이다.”

Organizing regional material-exchange markets and using them well is an important requirement for normalizing production at local-industry factories.



Conclusion



No previews in Pyongyang

Reform talk comes after the policy

Reform vocabulary rises about nine months after the state loosens the domestic economy, and it does not rise before.

Only for reform at home

Opening to foreign trade draws no response, and neither does re-centralization.

The result is robust

Three ways of measuring reform talk all find the same timing:

1. Breadth: distinct reform terms per article
2. Intensity: reform terms per length of text
3. The deliberately generous extended lexicon

THE FINDING

The journal explains and codifies reform **after the fact**, and only the reform that **needs justifying**.



What's next

A new method, still calibrating

An LLM helps discover and validate the reform terms, and a plain count of those terms in the text is what measures reform talk. It is a new way to do this kind of work. The pipeline, especially its LLM steps, is still calibrating.

The signal is strong, some details may change

These are initial results. We think the timing finding is right, but specific counts and term choices could shift as we iterate on the codebook and classification.

Look closer at the six-quarter echo

Reform talk rises most strongly three quarters after a domestic measure, and again, slightly weaker, at six quarters. Understanding what drives that second rise is next on the list.

Next: another round, then new sources

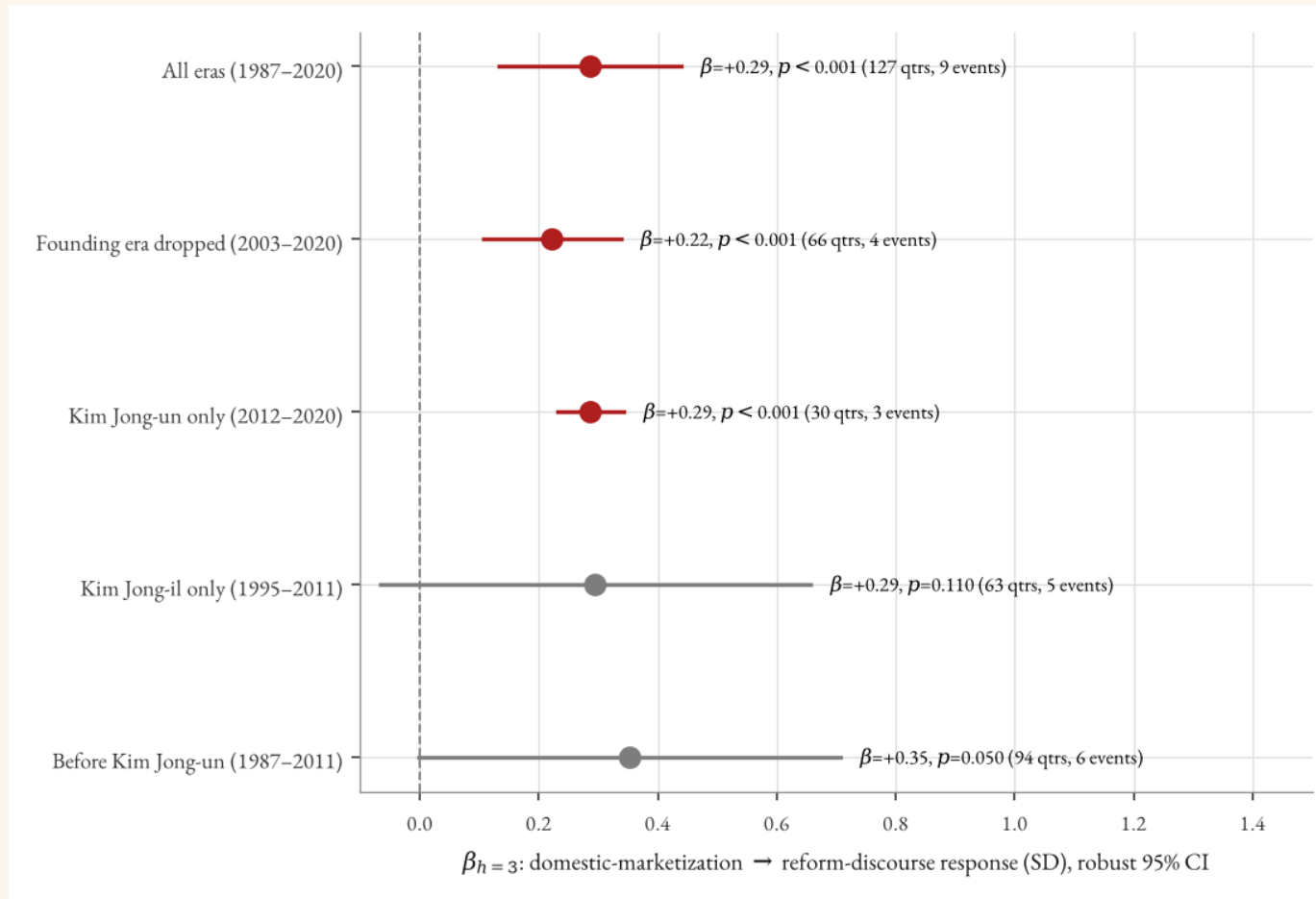
A further iteration of the coding and lexicon with expert review, then the same pipeline applied to other DPRK sources over time: KCNA, Rodong Sinmun, Chosun Munhak, and others.



Appendix

APPENDIX

Positive in every era



One estimate per subsample, all nested. Gray bars are wide because early issues are thin, not because the effect is weaker.

The other wiggles

$$\tau = -1$$

**a bump just before
a measure**

The one that would matter if it were real, because it would look like a preview. Its band includes zero, and the backwards test finds nothing at any horizon.

$$h = 6$$

**a smaller rise six
quarters out**

Not a second response. Two measures sit three quarters apart, so the sixth quarter after the first is the third quarter after the second. Drop the earlier measure and this rise halves while $h = 3$ does not move at all.

$$h = 8$$

a dip two years out

What a series looks like settling back after a temporary rise, not a policy effect running in reverse.

How a term enters the lexicon

1

Evidence pack

Corpus frequency,
reform-
discrimination lift,
era distribution,
usage snippets.
Judgment rests on
usage, not dictionary
sense.

2

Denotation judge

Does the term name
one of the six reform
mechanisms, or fall
under an exclusion?

3

Usage judge

Skeptical by
instruction.
Ambiguity goes to
the stricter bin.

4

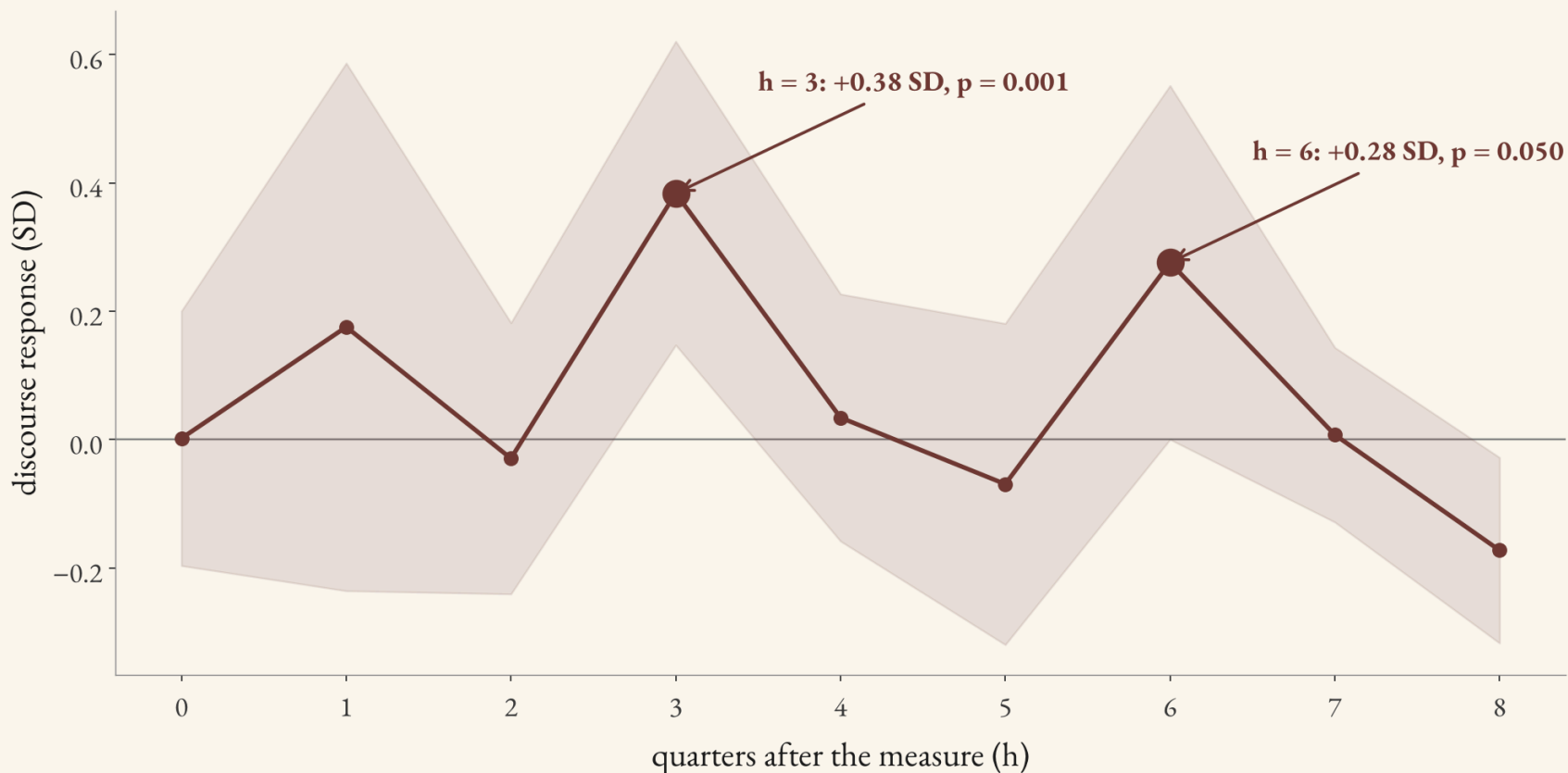
Arbiter

Burden of proof on
inclusion. Borderline
terms go to the
domain expert.

Of 32 candidates: 3 admitted, 11 to a robustness tier, 18 rejected.

APPENDIX

The marketization arm at $h = 6$



Same clustering as the aggregate series: dropping the 2001-10 measure alone nearly halves $h = 6$ ($+0.28$ to $+0.15$) while leaving $h = 3$ untouched ($+0.38$ to $+0.41$). Not an independent marketization-specific response.